Training Dataset

30 sentences

15 positive ✅ and 15 negative ❌ sentences used to train both models.

✅ POS	Good quality item, happy with the result.
✅ POS	Decent product, arrived on time, no complaints.
✅ POS	Pretty good overall, would buy again.
✅ POS	Nice product, works as described.
✅ POS	Satisfied with my purchase, does the job.
❌ NEG	This is absolute garbage, complete waste of money!
❌ NEG	Terrible product, broke after one day of use.

RNN — LSTM Model

`Embedding(vocab,16) → LSTM(32) → Dense(1,sigmoid)`

Epoch 20/20 — acc: 100.0% loss: 0.512

Dense — Bag-

`Input(vocabSize)`
`se(1,sigmoid)`

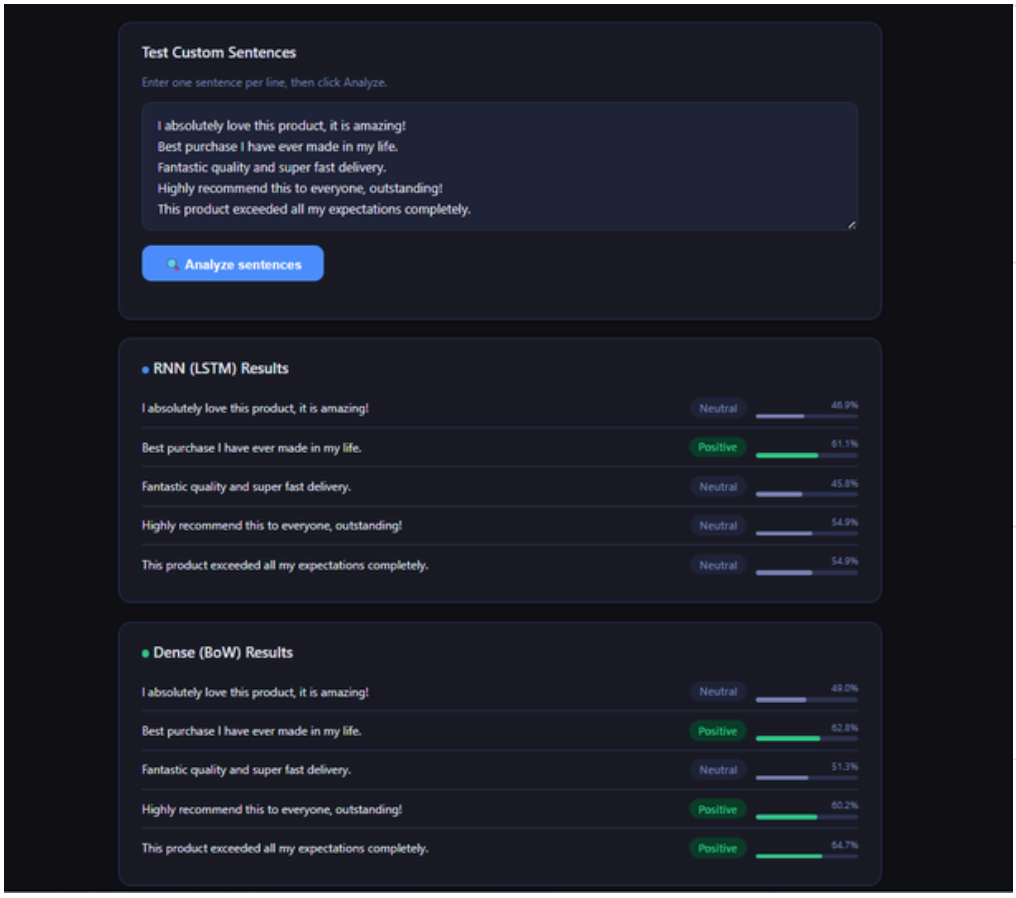
Epoch 20/20 — acc:

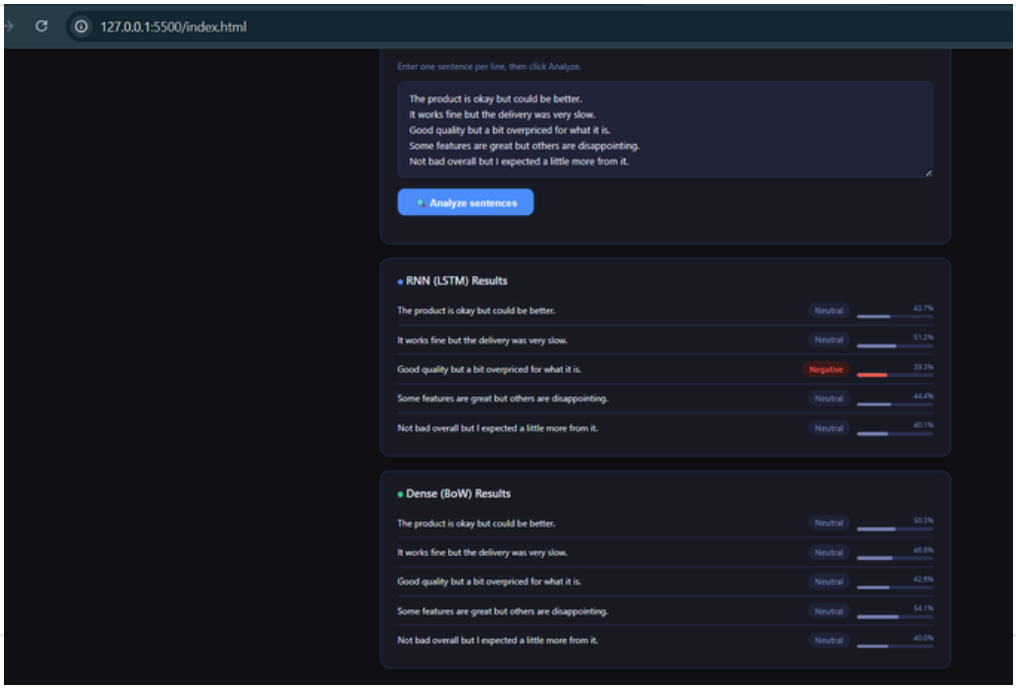
🔄 Retrain models

```
Dense epoch 15/20  acc=90.0%  loss=0.550
Dense epoch 20/20  acc=96.7%  loss=0.483

✓ Dense training complete!
RNN params:   9,505
Dense params: 6,465

🎉 Both models ready — switch to the Test or Compare tab!
```


Test Custom Sentences
Enter one sentence per line, then click Analyze.
I absolutely love this product, it is amazing!
Best purchase I have ever made in my life.
Fantastic quality and super fast delivery.
Highly recommend this to everyone, outstanding!
This product exceeded all my expectations completely.
Analyze sentences
RNN (LSTM) Results
I absolutely love this product, it is amazing! Neutral 46.9%
Best purchase I have ever made in my life. Positive 61.1%
Fantastic quality and super fast delivery. Neutral 45.8%
Highly recommend this to everyone, outstanding! Neutral 54.9%
This product exceeded all my expectations completely. Neutral 54.9%
Dense (BoW) Results
I absolutely love this product, it is amazing! Neutral 49.0%
Best purchase I have ever made in my life. Positive 62.8%
Fantastic quality and super fast delivery. Neutral 51.3%
Highly recommend this to everyone, outstanding! Positive 60.2%
This product exceeded all my expectations completely. Positive 64.7%


127.0.0.1:5500/index.html
Enter one sentence per line, then click Analyze.
The product is okay but could be better.
It works fine but the delivery was very slow.
Good quality but a bit overpriced for what it is.
Some features are great but others are disappointing.
Not bad overall but I expected a little more from it.
Analyze sentences
RNN (LSTM) Results
The product is okay but could be better. Neutral 43.7%
It works fine but the delivery was very slow. Neutral 51.2%
Good quality but a bit overpriced for what it is. Negative 39.3%
Some features are great but others are disappointing. Neutral 44.4%
Not bad overall but I expected a little more from it. Neutral 40.1%
Dense (BoW) Results
The product is okay but could be better. Neutral 50.3%
It works fine but the delivery was very slow. Neutral 46.6%
Good quality but a bit overpriced for what it is. Neutral 42.6%
Some features are great but others are disappointing. Neutral 54.1%
Not bad overall but I expected a little more from it. Neutral 40.0%

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Train | Test | Compare

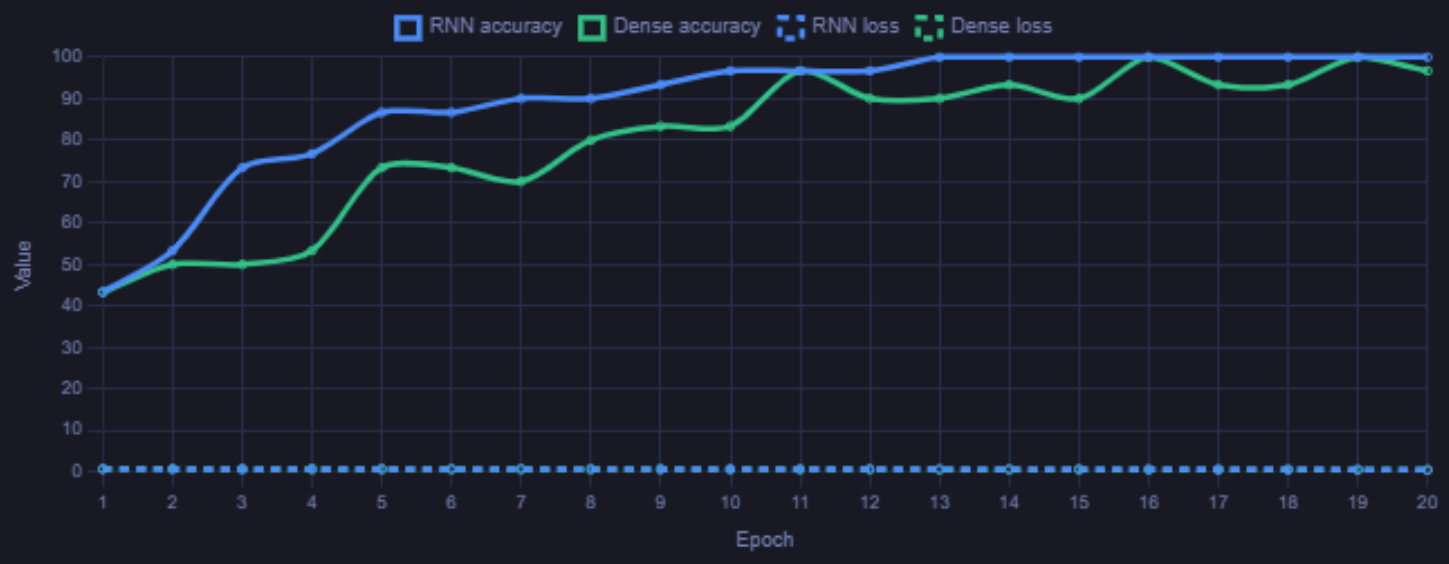
RNN — LSTM

Train Accuracy	Final Loss	Parameters
100.0%	0.512	9,505

Dense — BoW

Train Accuracy	Final Loss	Parameters
96.7%	0.483	6,465

Accuracy & Loss over Epochs



Architecture Analysis

RNN (LSTM): The embedding layer projects each word index into a dense 16-dimensional vector. The LSTM reads the sequence left-to-right while maintaining hidden state, so it can capture word order and context (e.g. "not bad" vs "bad"). This makes it stronger for nuanced, longer sentences.

Dense (Bag-of-Words): Converts the sentence to a binary vector where each slot = "word present/absent". The dense layer learns weighted combinations of word signals. Word order is lost, but the model trains faster and is highly stable on small datasets.

Small dataset insight: With only 30 sentences both models can overfit quickly. The Dense BoW model typically converges faster due to fewer parameters. The LSTM model benefits more from larger corpora where sequence-level reasoning matters. On production datasets (10k+ samples) the LSTM advantage becomes clear.